

Optimizing Cooperative Perception in Autonomous Vehicle through Semantic Data Analytics

Presented By:

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Overview – Problem Statement

Jia Xuan Goh

Every vehicle sees through **one pair of eyes**.

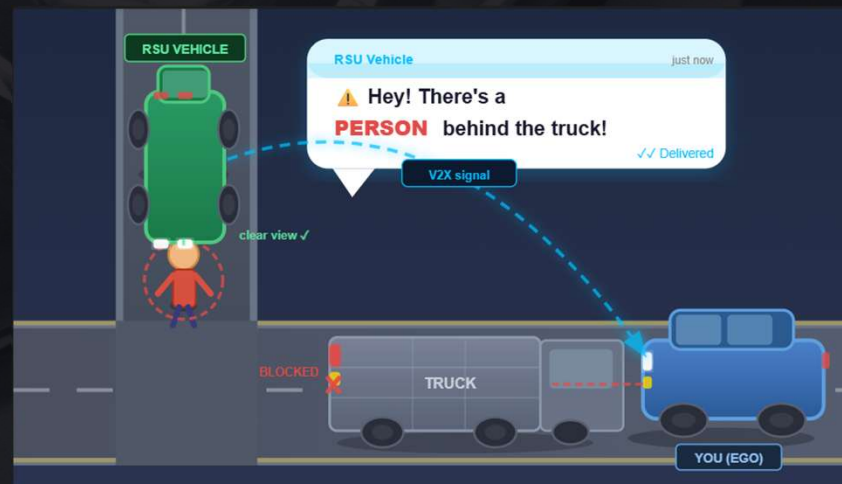
- Roads are shared. Blind spot aren't.
- Trucks, buildings– object hide constantly.

What if cars could **talk/share information**?

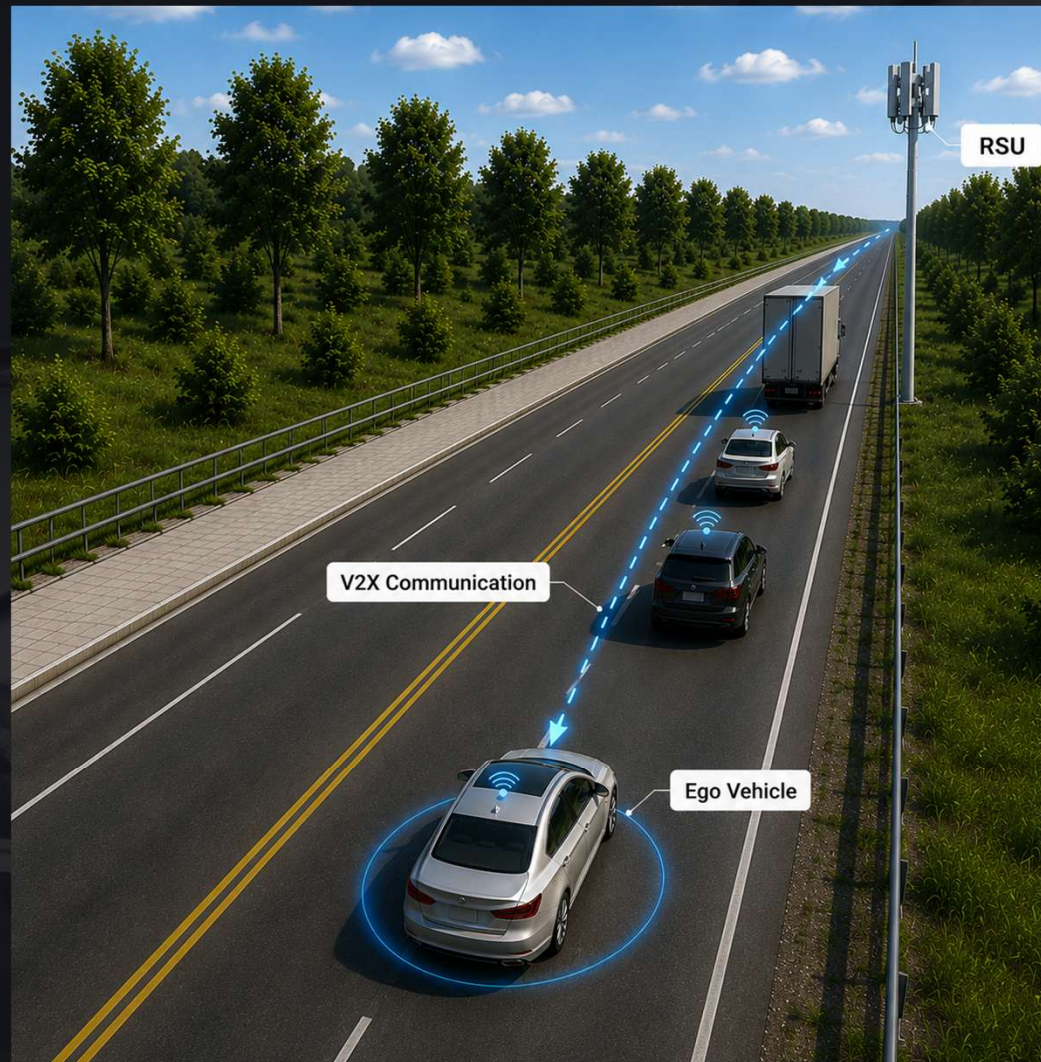
- We already do this as humans – WhatsApp, phone calls
- V2X is not a research fantasy – it's an international standard
- Every camera-equipped car become a cooperative node

how important is it?
how smartly they do it?

Road Side Unit (RSU)



Ego



Project Objective – Goals, Scope

Jia Xuan Goh

R1: Baseline Detection

- Detect vehicles, people and objects from camera
- Establishes the “before cooperative” benchmark

R2-R3: Cooperative Perception

- R2: RSU reveals hidden vehicles. Ego identified what they are
- R3: RSU only contacted when confidence drop below threshold

R4: Minimise RSU transmission payload

- Minimum JPEG quality that preserves detection accuracy
- Skip the image — send detection result only

Out of scope

- Latency performance
- Real-time video streaming

Research Basis

- Cooperative perception literature
- When · What · How little



Project Objective – User Group, Use Cases

Pei Shuen Ooi

User Group

Primary

AV Developers

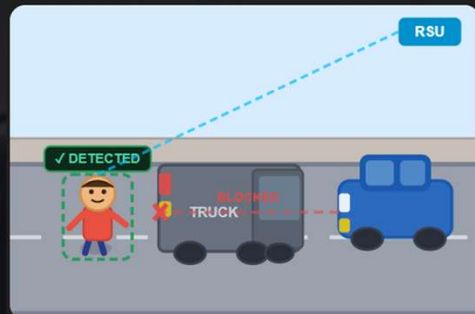
End User

Road Users

Use Cases



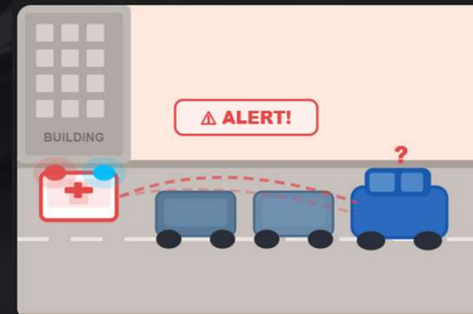
Occluded detection



Blind intersection

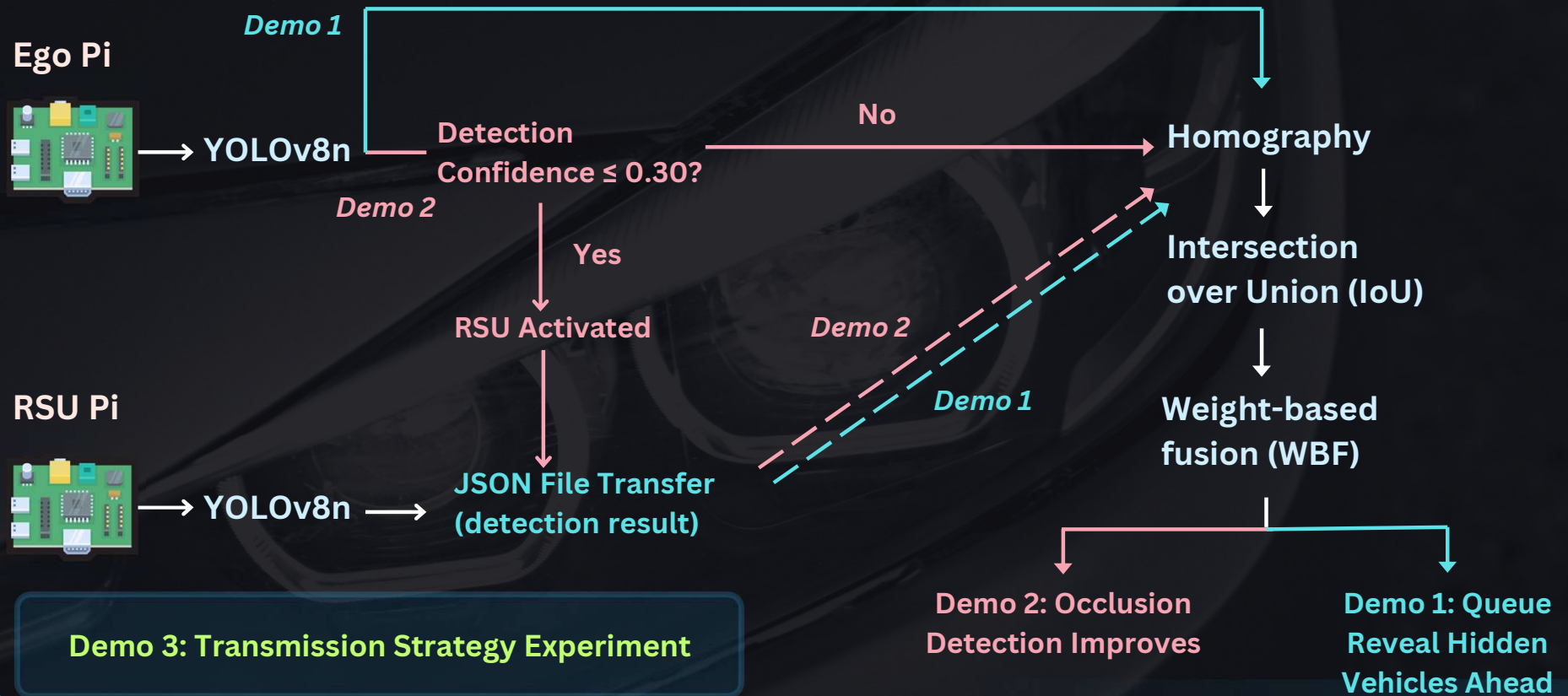


Emergency Vehicle



Project Deliverables – System Architecture

Pei Shuen Ooi



Project Deliverables

Pei Shuen Ooi

Hardware

➞ Raspberry Pi

➞ Pi Camera

➞ Mobile Hotspot

➞ Laptop

Software

➞ Raspberry Pi OS

➞ Raspberry Pi Connect

➞ VS Code

➞ SSH

Tech Stack

➞ Python

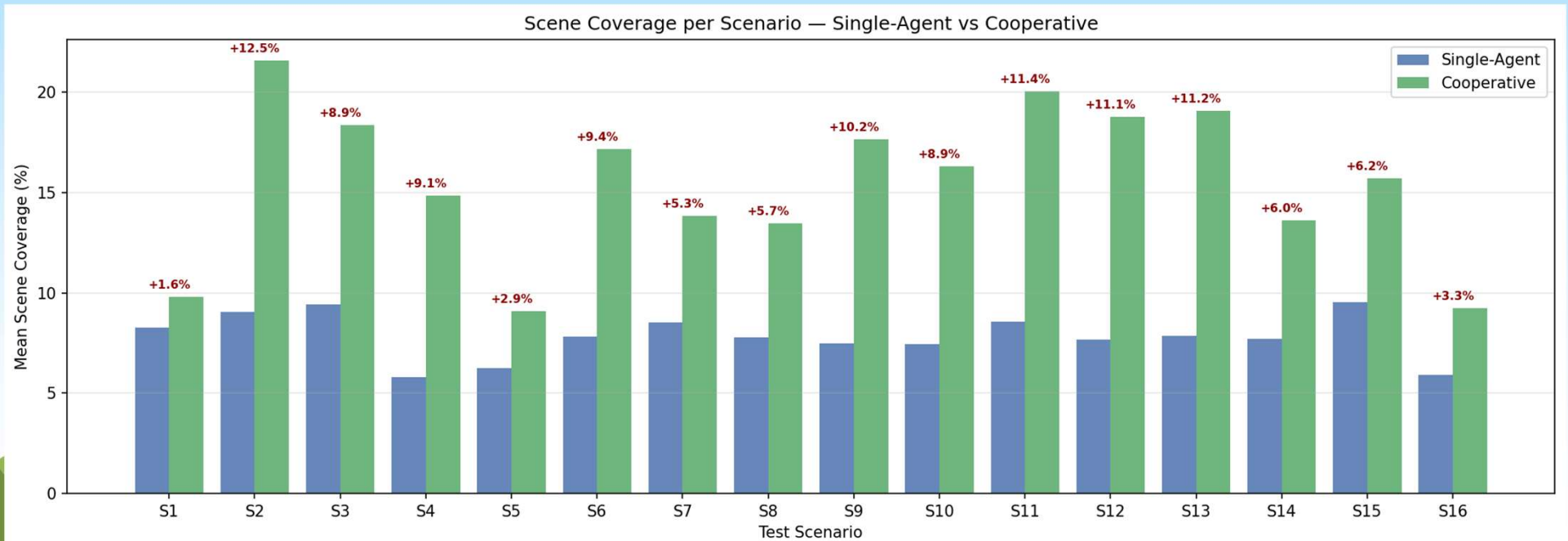
➞ YOLOv8n

➞ OpenCV

➞ Weighted Box Fusion

Result:
Scene Coverage on OPV2V (R1)

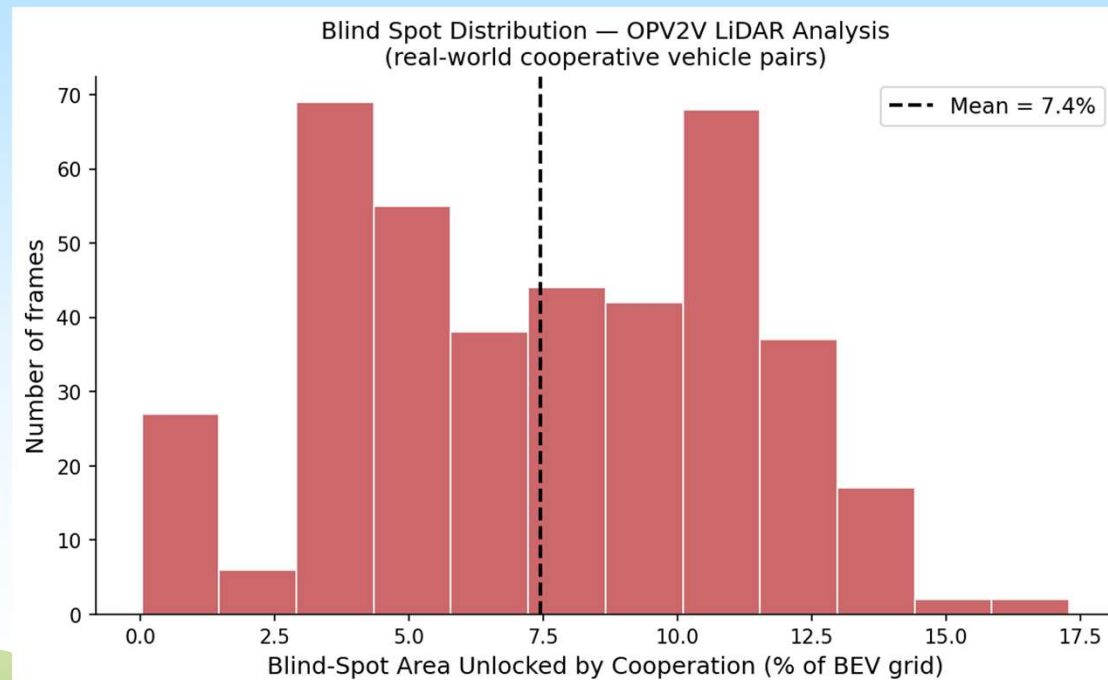
Result: Scene Coverage



Result:

Blind Spot Elimination on OPV2V (R1)

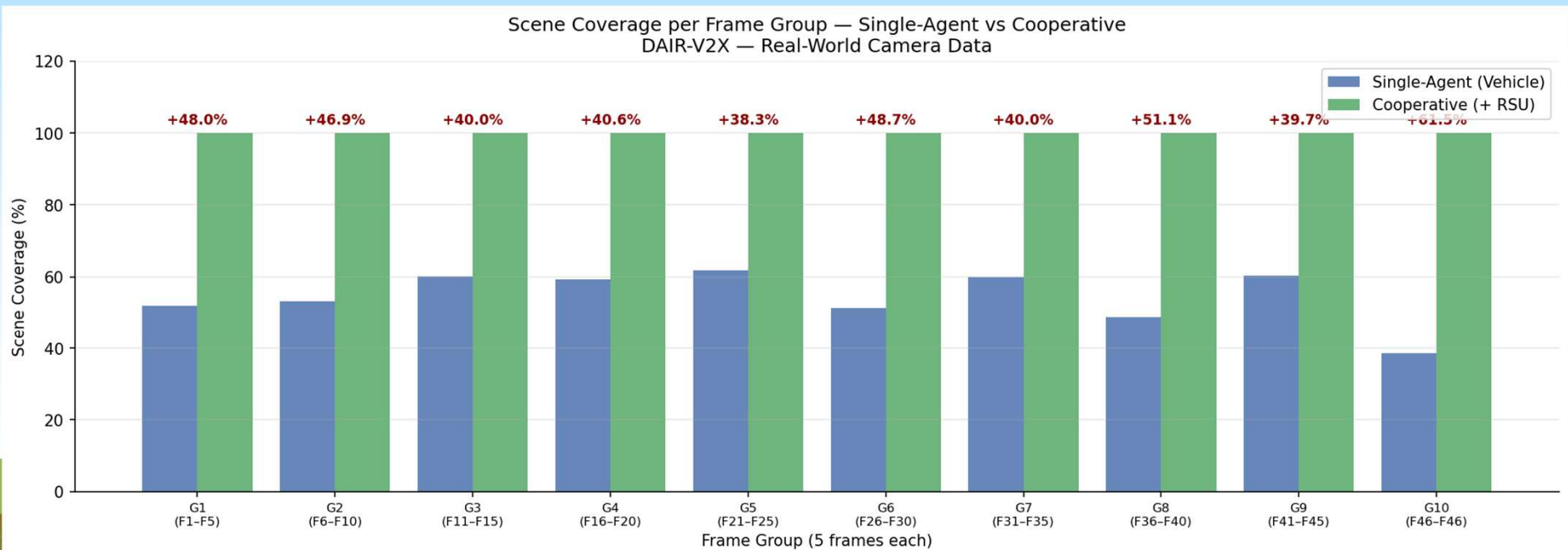
Result: Blind Spot Elimination



Result:

Scene Coverage on DAIR-V2X (R1)

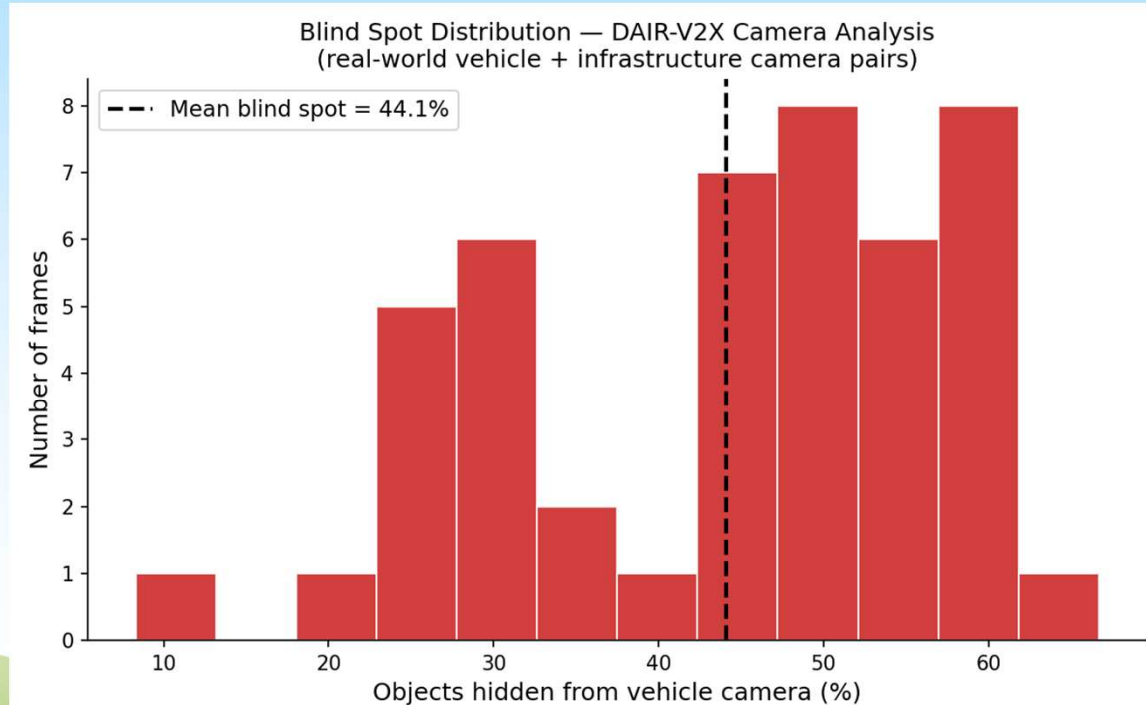
Result: Scene Coverage



Result:

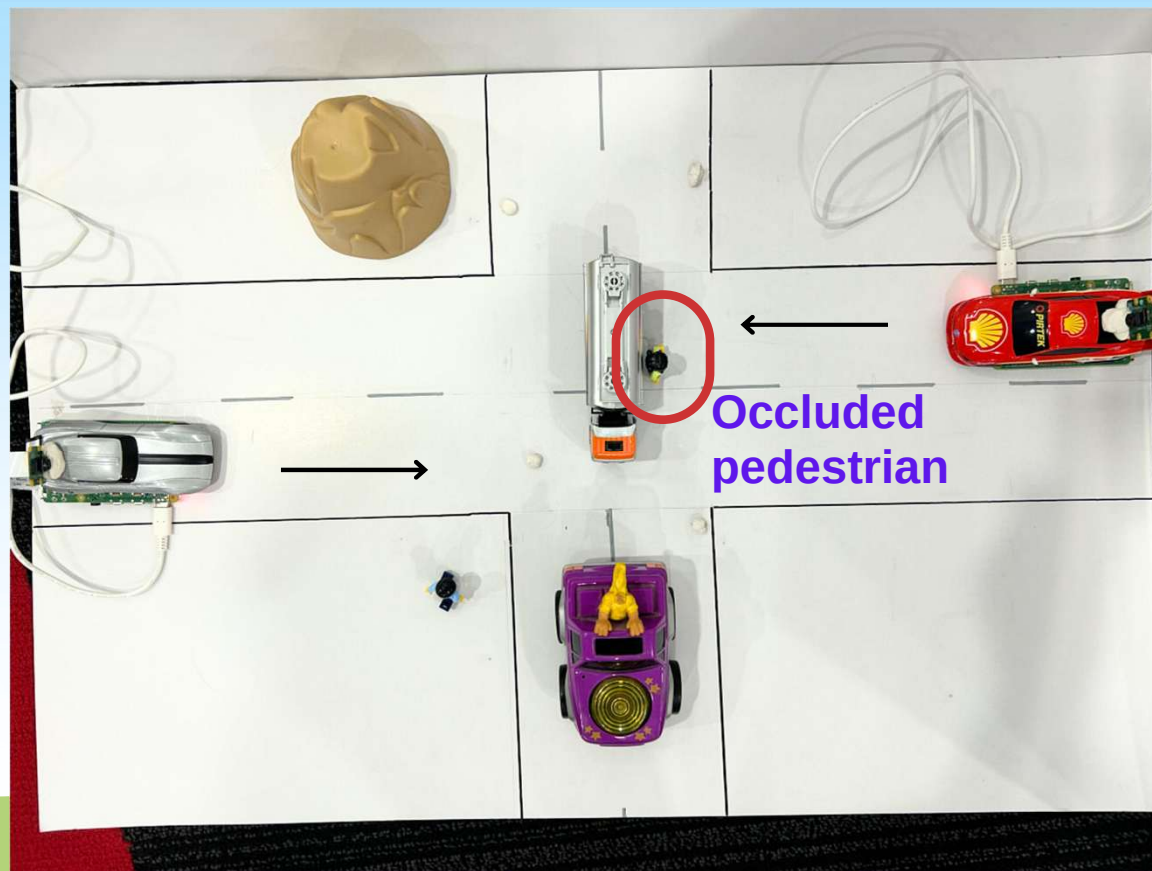
Blind Spot Distribution on DAIR-V2X (R1)

Result: Blind Spot Distribution



Demo

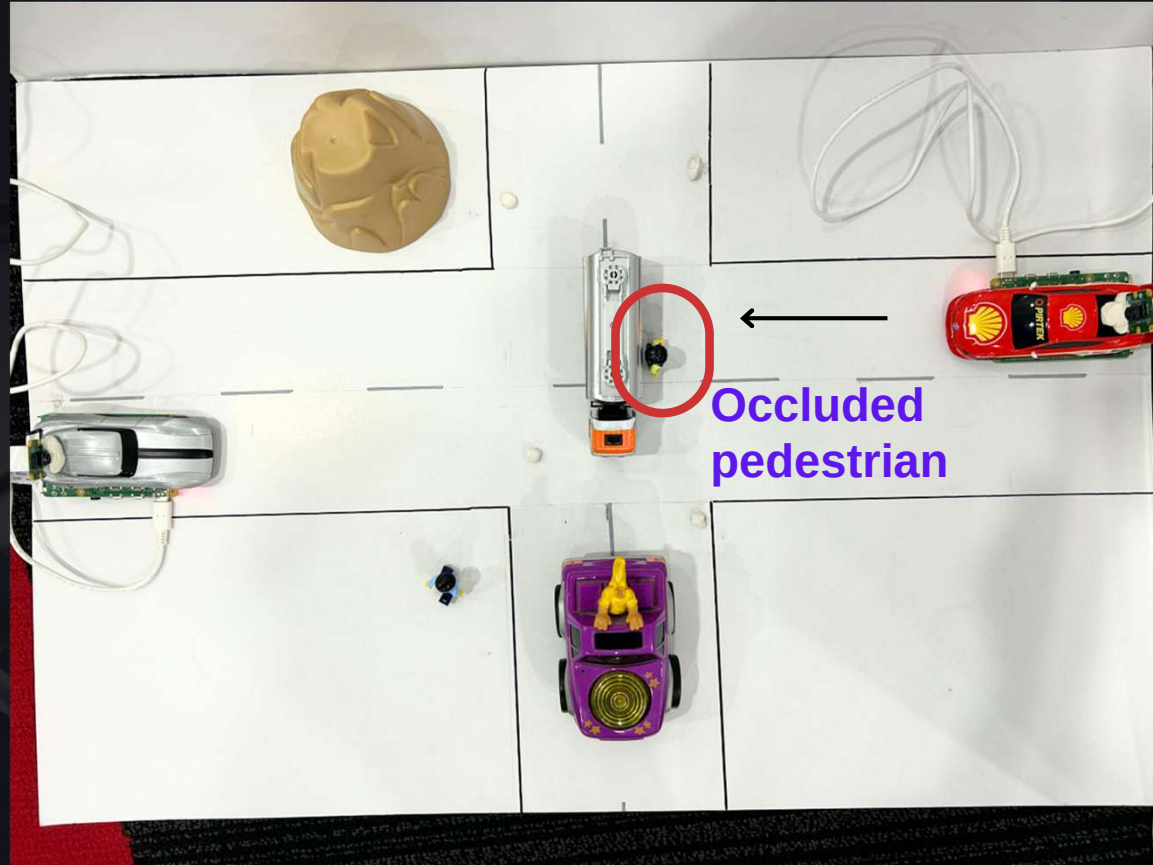
Ego
(Pi 1)



V2
(Pi 2)

Occluded
pedestrian

Ego
(Pi 1)



V2
(Pi 2)

Occluded
pedestrian



Demo (R2)

Panel A ??? Ego Camera (Single Vehicle)



Ego vehicle

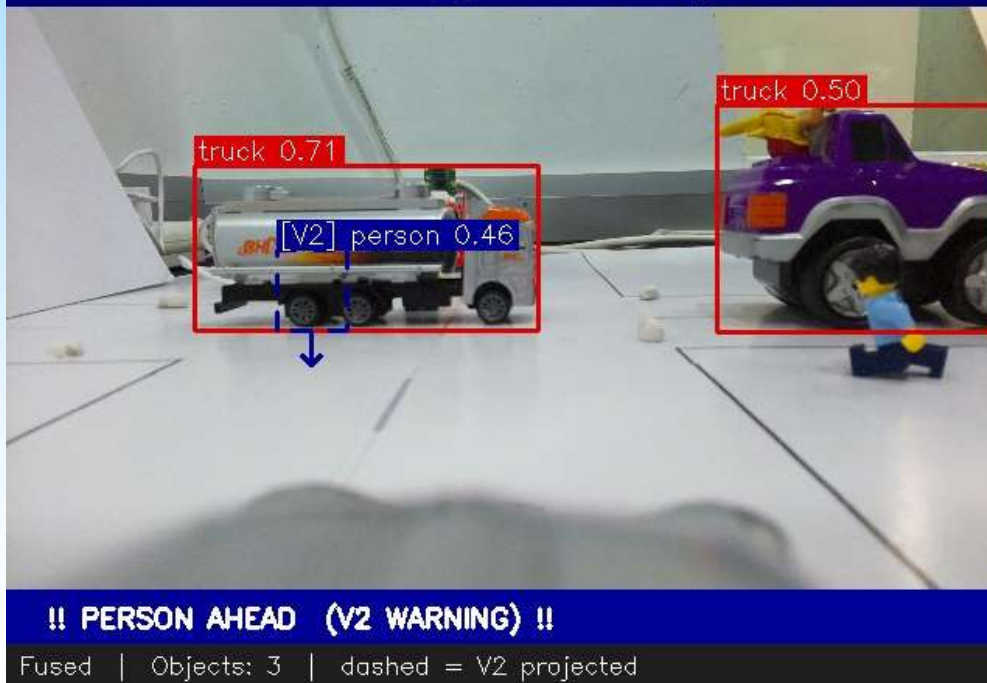
Panel B ??? Vehicle 2 Camera (Cooperative Partner)



Spectator vehicle

Demo (R3)

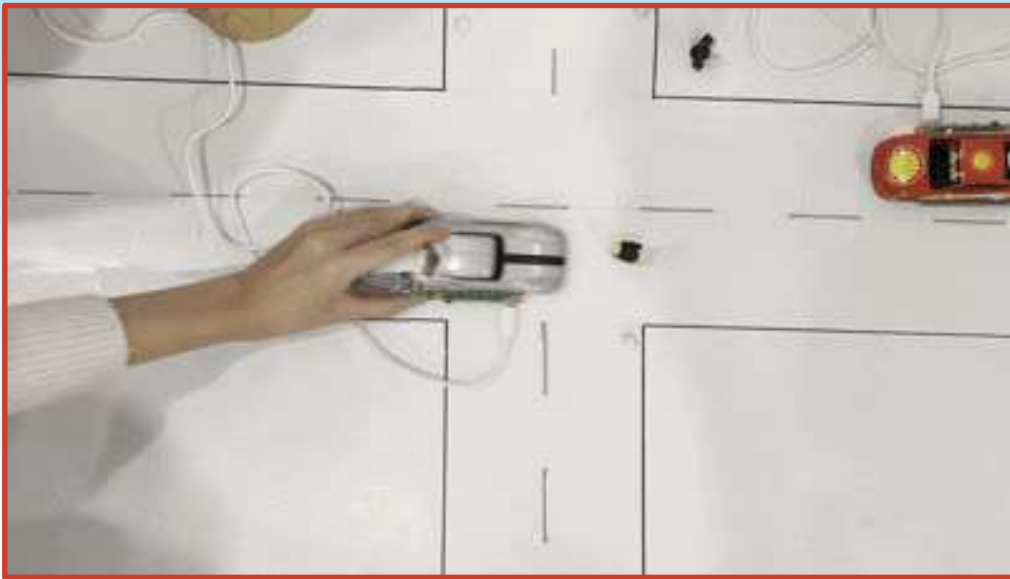
Panel C ??? Fused View (Ego + Vehicle 2)



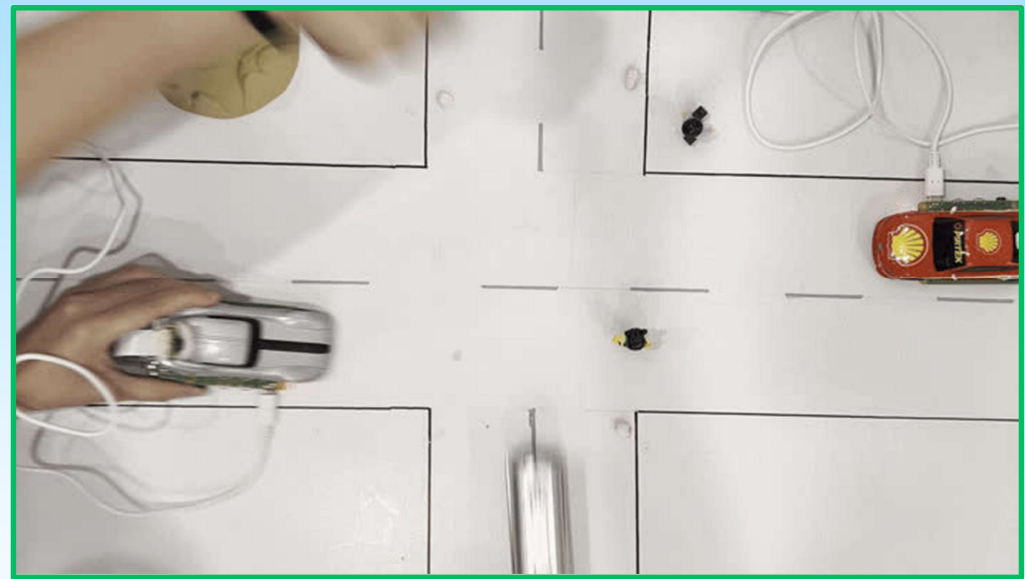
Fused cooperative perception

The ego vehicle sees an occluded pedestrian behind the truck

Demo



**Without cooperative
perception**



**With cooperative
perception**

Jia Xuan Goh

Demo 1

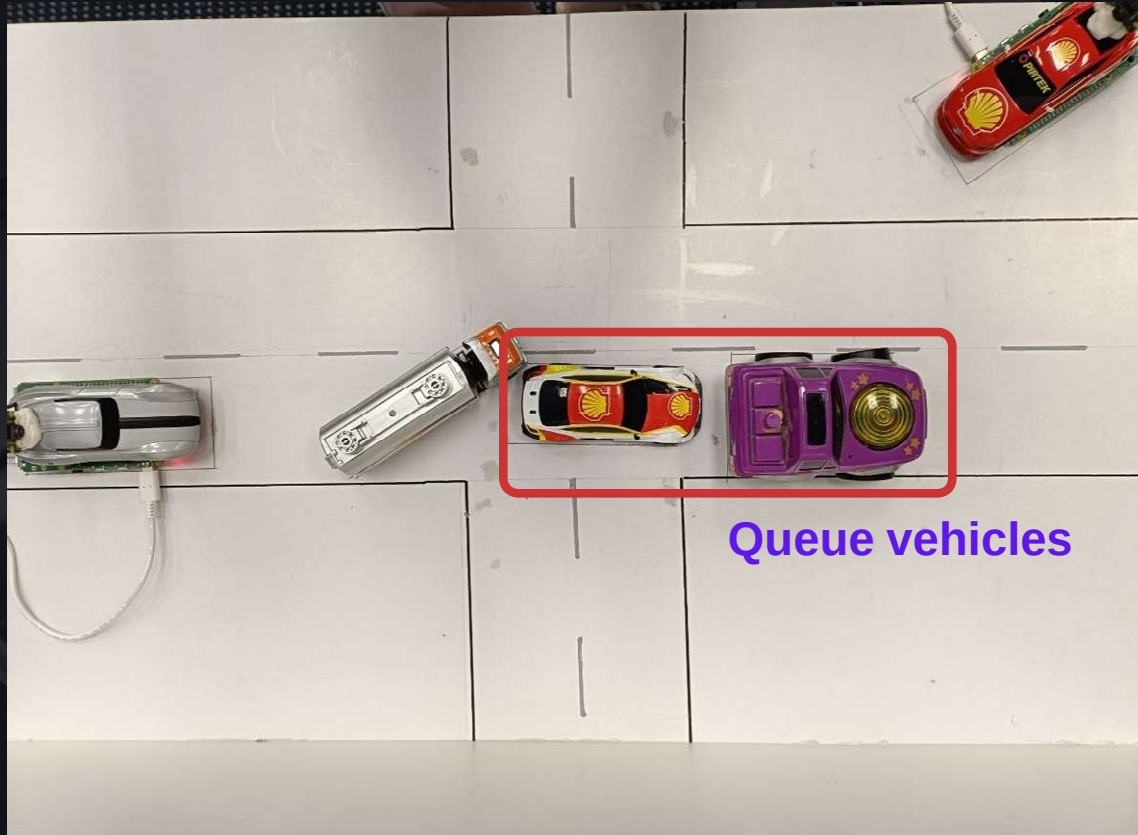
Seeing Beyond the Truck: Cooperative Queue Perception



Demo 1 Setup

Jia Xuan Goh

Ego
(Pi 1)

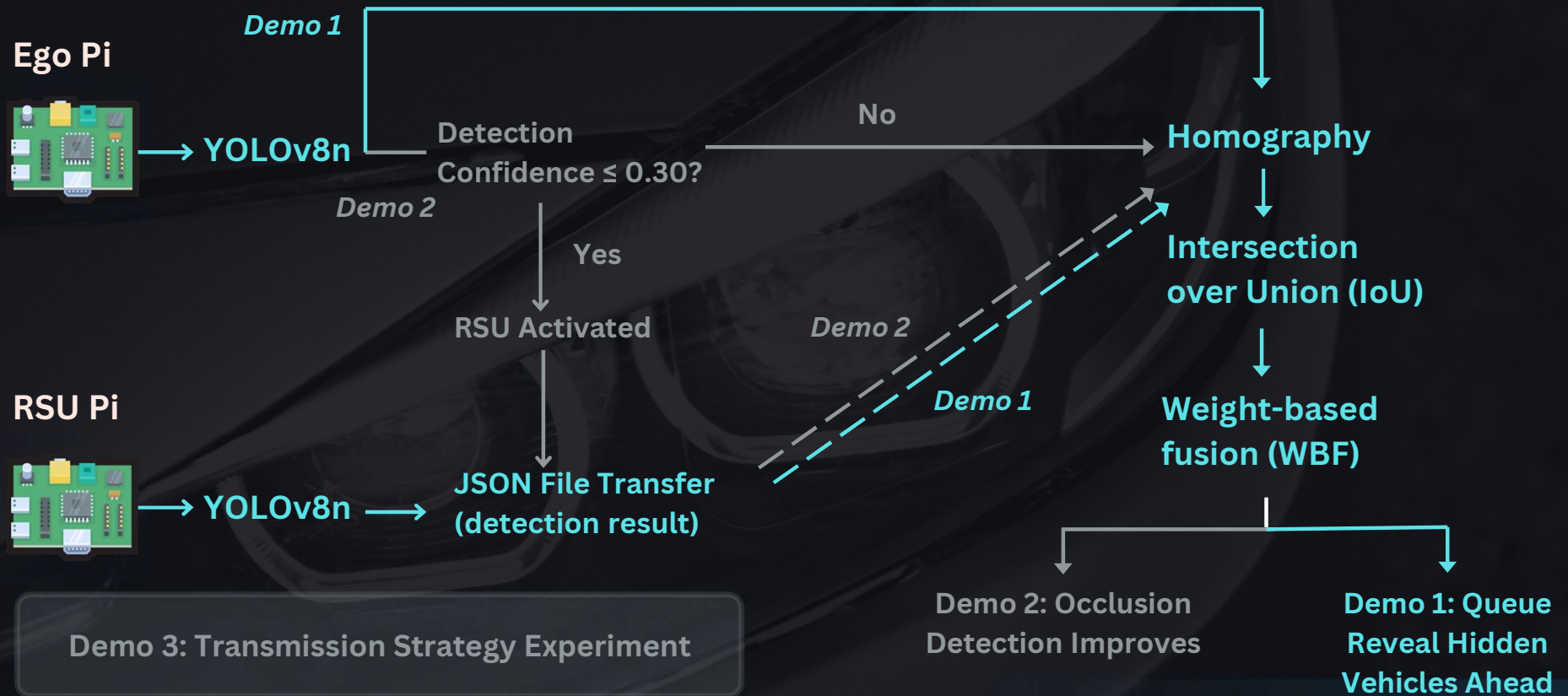


RSU
(Pi 2)

Queue vehicles

Project Deliverables – System Architecture

Pei Shuen Ooi



Fusion Pipeline

Jia Xuan Goh

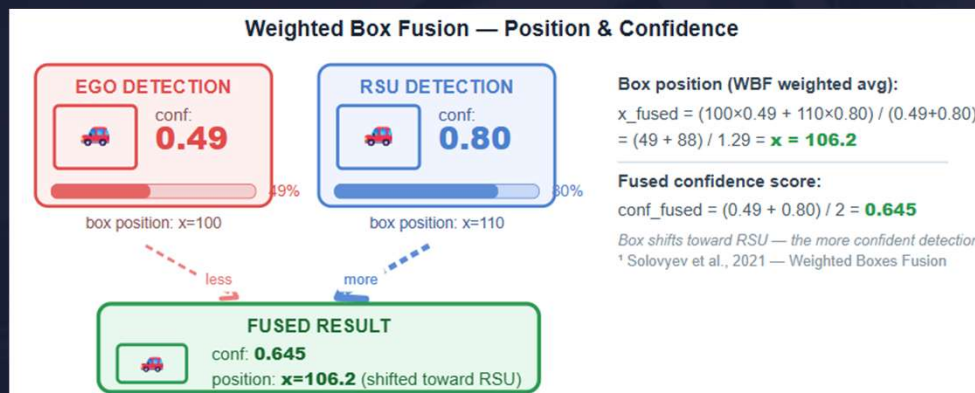
Homography - Coordinate Projection

Shared anchor maps RSU detections into ego's coordinate space

IoU - Match or Hidden

Checks overlap between bounding boxes

Weighted Box Fusion (WBF)



Box position weighted by confidence
— higher confidence shifts the final box position more

¹ (Solovyev et al., 2021 — Weighted Boxes Fusion)

Jia Xuan Goh

Demo 1 Begin

Seeing Beyond the Truck:
Cooperative Queue Perception



From Data to **Decision**

Jia Xuan Goh



Queue lane selection

Approaching a junction —
which lane is shorter?



Overtake or reroute

On a single-lane road —
is it worth waiting?



Demo 2

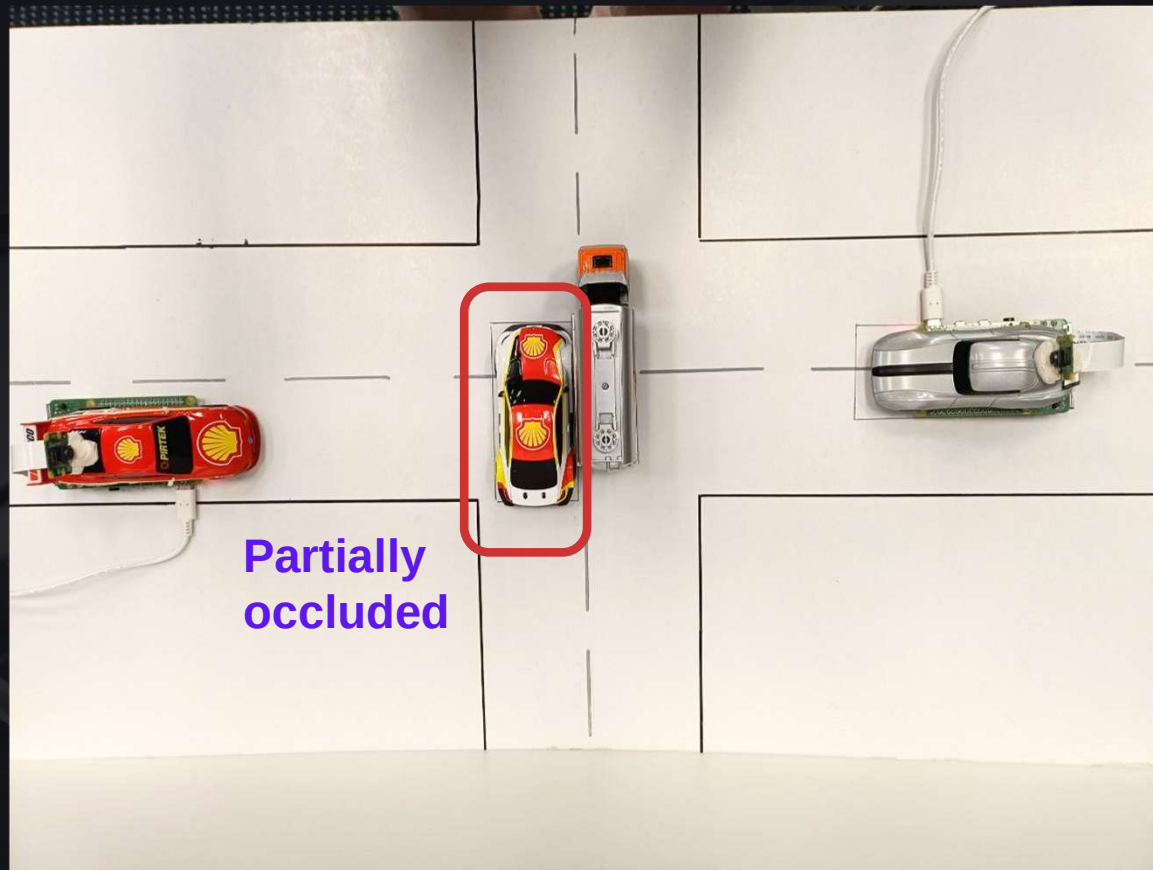
Seeing Through Occlusion:
Confidence-Triggered Cooperative Perception



Demo 2 Setup

Pei Shuen Ooi

RSU
(Pi 2)

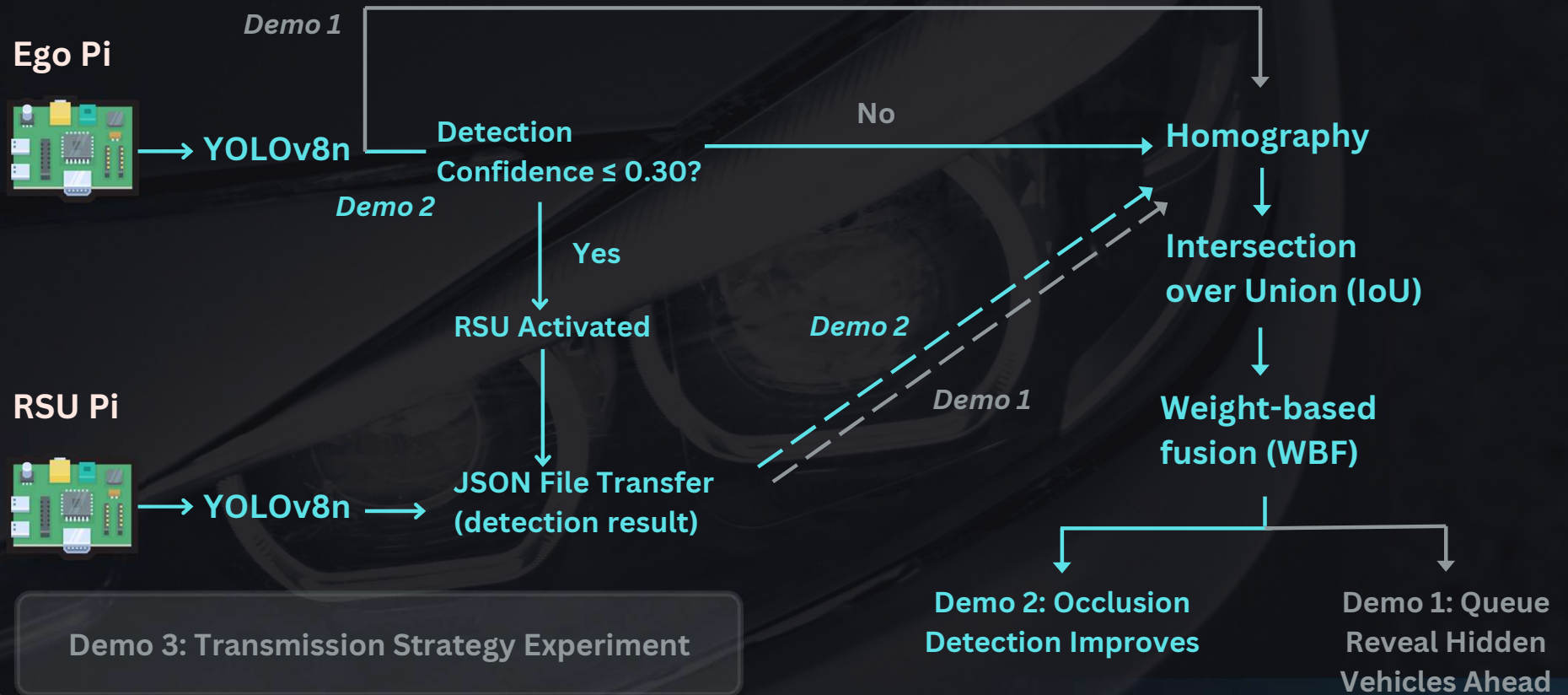


Partially
occluded

Ego
(Pi 1)

Project Deliverables – System Architecture

Pei Shuen Ooi



Demo 2 Begin

Seeing Through Occlusion:
Confidence-Triggered Cooperative Perception



From Data to **Decision**

Pei Shuen Ooi



React before it's too late

High occlusion blocks the AV's view – RSU reveals the hidden car so it stops in time



Reduce communication cost

RSU only activates when needed – scalable for real-world deployment



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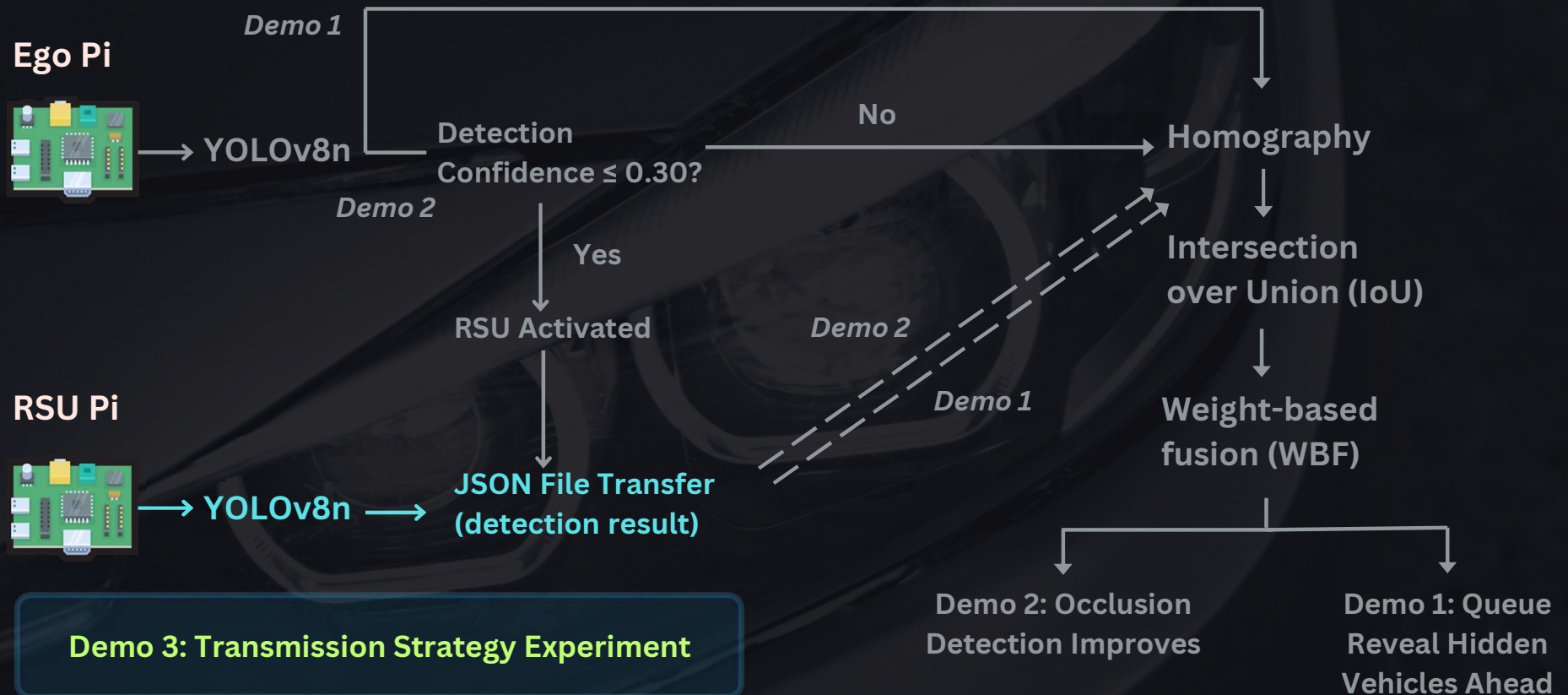
Demo 3

From Raw Image to Detection Result



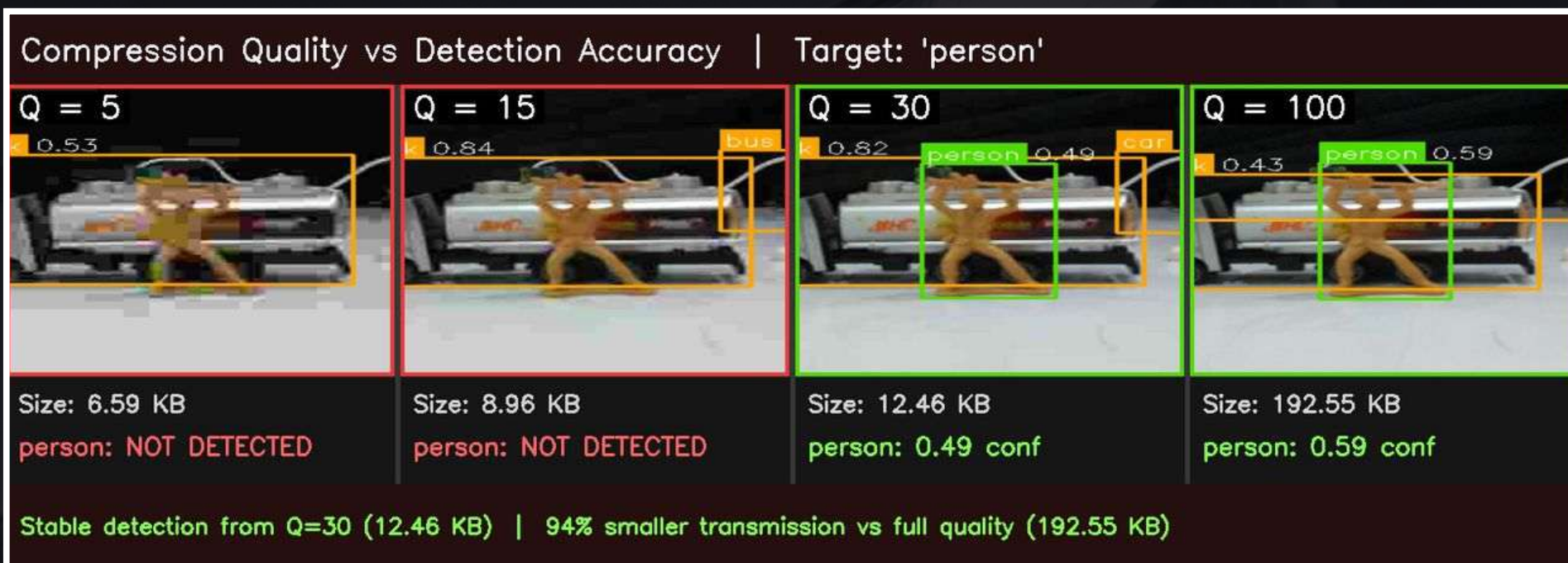
Project Deliverables – System Architecture

Pei Shuen Ooi



How Much Can We **Compress**?

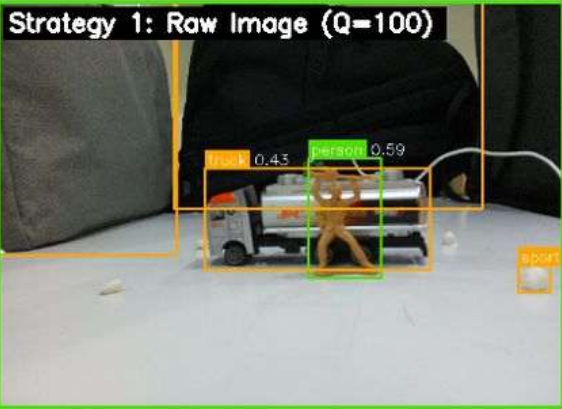
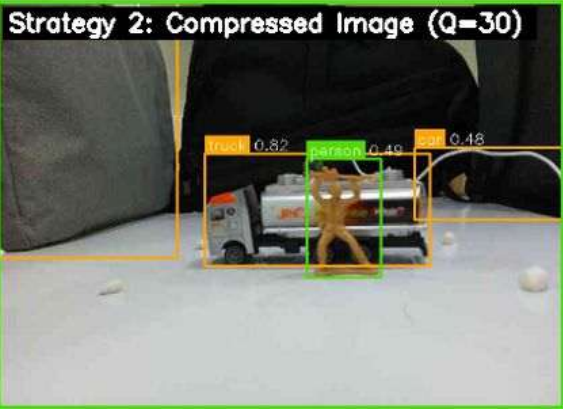
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Q = 30 is our **minimum viable quality** for object detection

What If RSU **Sends Results** Instead?

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RSU Transmission Strategy Comparison Target: 'person'		
Strategy 1: Raw Image (Q=100) 	Strategy 2: Compressed Image (Q=30) 	Strategy 3: Semantic Only <pre>[{ "class": "person", "conf": 0.595, "bbox": [351, 186, 432,</pre>
Payload: 192.55 KB person: DETECTED Confidence: 0.59	Payload: 12.46 KB person: DETECTED Confidence: 0.49	Payload: 0.0645 KB person: DETECTED Confidence: 0.60
Semantic payload (0.0645 KB) is 100.0% smaller than raw image (192.55 KB)		

Up to **99.9% less data transmitted** from raw to semantic

Project **Deficiencies**

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Bounding Box Drift

Hidden vehicles positions are estimates – drift increases from anchor



Fixed Confidence Threshold

Threshold is not adaptive to different environments

Every deficiency points to a clear next step

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Conclusion



Lesson Learnt

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Effective Teamwork

Unsustainable workload imbalance

Each member owns and presents one demo end-to-end



Effective Project Management

Original plan too ambitious on hardware

Scoped down – focused on delivering validated results

Recognise the imbalance early, and prioritise what delivers – not what impresses.



Future of project

Li Xian Ng



Multi-RSU & Live streaming

Ego communicates with multiple RSUs · static frames → live video stream



Latency performance

Benchmark end-to-end delay · define a real-time threshold



Li Xian Ng

Thank You 